2E-EDUC 6- FS Field Study

Term 2 AY 2025 - 2026

Worksheet on Writing a Classroom-based Action Research (CBAR) Concept Paper/Proposal

Name: EVARDO, Ezekiel John M.
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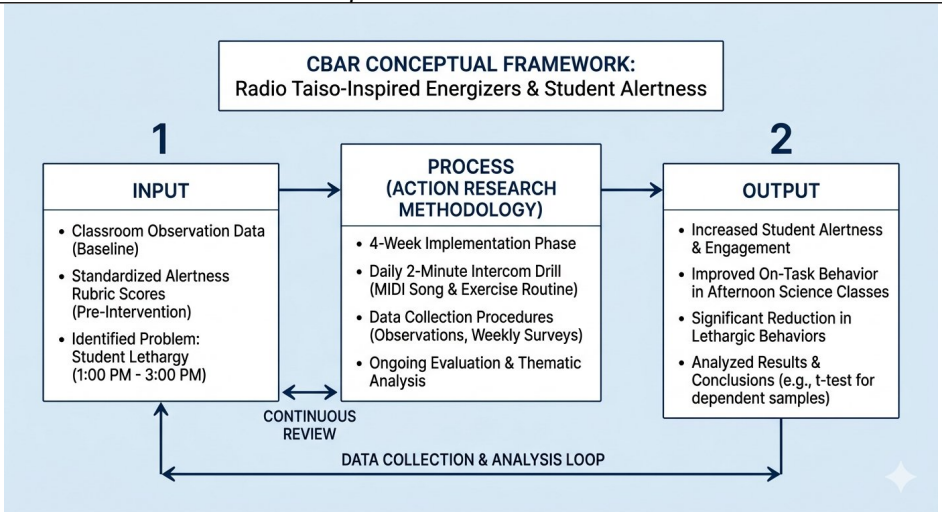
Direction:

- After familiarizing yourself with the various sections/elements of an action research report as well as gaining skills in critiquing of research report, you are now ready to write your CBAR proposal/ concept paper.
- Formulate one title from your observations and experiences in all the professional education courses that you have taken especially from your Field Study course, and write your CBAR proposal/concept paper following the format below
- **You may write your CBAR proposal without the table following the guide questions in each section.**
- Then, do self-assessment of your CBAR proposal/concept paper by accomplishing the assessment rubrics found below.

CBAR PROPOSAL TITLE/TOPIC: Note: A good tile should contain topic, action, and intervention.	Radio Taiso-Inspired Energizers: Enhancing Student Alertness and Engagement during the Afternoon Slump A Concept Paper/Proposal
NAME OF PROPONENT:	EVARDO, Ezekiel John M/

I.	**Rationale (Brief Description of the study)**	• *What is the research about (in general)?*
		  o *This study investigates the impact of a structured, two-minute musical exercise routine—inspired by the Japanese Radio Taiso—conducted over the school intercom to combat student lethargy randomly during the 1:00 PM – 3:00 PM period.*
		• *What is/are the trigger(s) for this action research? What is the major issue to be studied?*
		  o *The "afternoon slump" is a significant barrier to learning in Science education, where students frequently demonstrate sedentary behavior, reduced focus, and decreased participation.*
		• *Include review of related literature/studies*
		
		Recent educational research indicates that classroom-based physical activity interventions show promising results for improving student alertness and engagement, with consistent positive effects on on-task behavior across multiple studies. The evidence is particularly strong regarding the reduction of sedentary behavior during instructional time. For instance, Mahar et al. (2006) demonstrated an 8% overall improvement in on-task behavior among 243 students, noting that the least engaged students showed a significant 20% increase in focus following physical activity. Similarly, Goh et al. (2016) reported that on-task behavior increased from 82.3% to 89.5% across nine different classes following brief active breaks. These findings are supported by Carlson et al. (2015), whose study of 1,322 students found that the implementation of physical movement was negatively associated with a lack of effort and off-task behavior.
		
		Regarding the specific cognitive mechanisms of these "energizers," researchers suggest that the nature of the activity matters as much as the movement itself. Schmidt et al. (2016) found that cognitive engagement—rather than physical exertion alone—enhanced focused attention in children. This supports the use of structured, memorized routines like the Japanese-inspired drills, which require students to recall specific movements in sync with music. A systematic review by Ruhland et al. (2021) concluded that the available evidence points to a beneficial effect of exercise on both attention and on-task behavior, reinforcing the rationale for using brief drills to mitigate the "afternoon slump".

II.	**Significance of the study to the research priorities of DepEd / your Institution**	*Indicate the significance of the research study findings to various levels / specific personnel /target groups in the organization (such as supervisors/principals, teachers, students, parents, etc.)* • *To Students: Provides a physiological "reset" to improve focus and academic performance during late-session classes.* • *To Teachers: Offers a standardized, time-efficient classroom management tool to regain student attention without consuming instructional time.* • *To School Administration: Promotes a culture of wellness and provides a scalable model for school-*

	wide engagement via existing infrastructure (intercoms).
• **Statement of Purpose/Objectives**	• *Write a statement that conveys the overall objective or intent of your action research.* 1. *To evaluate the effectiveness of a daily 2-minute intercom-led exercise routine in reducing student lethargy and increasing engagement in afternoon classes.* • *List the research questions/objectives that you plan to answer/reach to achieve the purpose of your action research.* 1. *What is the level of student engagement during afternoon Science classes before and after the implementation of the exercise drill?* 2. *Is there a significant difference in student alertness levels after the 2-minute routine?* 3. *How do students perceive the influence of the musical drill on their readiness to learn?*
IV. Graphical Conceptual Framework	
V. Methodology	*This Classroom-Based Action Research (CBAR) utilizes a quasi-experimental design to evaluate the impact of a daily, 2-minute intercom-led physical exercise routine on reducing student lethargy during the afternoon slump. Data will be gathered through pre- and post-intervention observations using a standardized alertness rubric and weekly student surveys to measure shifts in engagement. Quantitative results will be analyzed using a t-test for dependent samples to determine statistical significance, while qualitative feedback will undergo thematic analysis to contextualize the students' experiences.*
A. Type of Data Needed	• *Describe data that are needed for the study (i.e. survey, interview, observation, literature analysis, validation rubrics, etc.)* a. *Quantitative Data: Numeric scores derived from structured observation checklists to measure*

	frequency and intensity of on-task behaviors and physical alertness during the 1:00 PM – 3:00 PM window. *b. Qualitative Data: Descriptive narratives from open-ended survey responses and teacher field notes to capture student perceptions, mood shifts, and any barriers to participation.* *c. Triangulation: The use of both data types allows for a more robust analysis of how the 2-minute drill impacts the classroom environment.*
B. Research Instrument for Gathering Data	• *Describe tools/instruments that will be used In reference to the types of data needed* *a. Standardized Alertness Rubric: A 5-point Likert scale (1 = Lethargic/Non-responsive to 5 = Fully Alert/Engaged) used by the researcher to pulse-check the room 10 minutes before and 10 minutes after the intercom drill.* *i. 5 (Highly Alert) - Students are upright, eyes focused on the teacher/task, and respond immediately to verbal cues.* *ii. 4 (Attentive) - Students are mostly upright; occasional shifting, but generally focused on instructional materials.* *iii. 3 (Passive) - Students appear "blank"; minimal movement, slow response time, but no visible signs of sleeping.* *iv. 2 (Lethargic) - Visible slouching, head-propping (leaning on hands), frequent yawning, and slow to follow directions.* *v. 1 (Drowsy) - Heads on desks, closed eyes, or complete lack of engagement with the lesson environment.* *b. Post-Intervention Survey: A brief, 3-item questionnaire administered weekly. It includes two closed-ended questions on energy levels and one open-ended prompt: "How did the musical routine affect your readiness for the upcoming lesson?".* *i. How would you rate your energy level immediately after the 2-minute musical drill?* *1. (1) Still very tired (2) Slightly more awake (3) Moderate energy (4) Energetic (5) Very refreshed* *ii. Did the music and exercise help you focus better on the Science lesson that followed?* *1. (1) Not at all (2) A little bit (3) Somewhat (4) Mostly (5) Definitely* *iii. In your own words, how did the musical routine affect your readiness to learn today?* *c. Behavioral Checklist: A tally sheet to record specific indicators of the "afternoon slump," such as head-propping, yawning, or off-task conversation, to be compared against baseline data.* *i. Head-propping: (Leaning chin or side of head on hands/arms)* *ii. Off-task Conversation: (Whispering/talking about non-lesson topics)* *iii. Physical Slumping: (Sliding down in chair or resting head on desk)* *iv. Yawning/Stretching: (Overt signs of physical tiredness)*

	v. Delayed Start: (Taking more than 30 seconds to open materials/notebooks)
C. Sample Needed/Respondents	• *Who are the participants (i.e. students, teachers, principals, experts, etc.). Describe their involvement.* *a. One section of students (approx. 40-50) attending classes between 1:00 PM and 3:00 PM.*
D. Data Collection Procedure	• *List and describe steps or procedures* *Phase 1: Baseline (Week 1): Observation of students using the Behavioral Checklist and Alertness Rubric without the intervention to establish "normal" lethargy levels.* *Phase 2: Instructional (Week 2): Formal orientation where students memorize the 2-minute routine and become familiar with the MIDI audio cues played via the intercom.* *Phase 3: Implementation (Weeks 3–6): Daily execution of the drill; the researcher records alertness scores 10 minutes post-drill and collects survey responses every Friday.* *Phase 4: Finalization (Week 7): Consolidation of observation logs and survey data for comparative analysis.*
E. Data Analysis Method	• *Use of Descriptive Statistics (mean scores of alertness) and Thematic Analysis for student feedback.* *a. Descriptive Statistics: Calculation of the Mean ($\bar{x}$) and Standard Deviation (σ) for alertness scores to summarize the central tendency and spread of student engagement during both the Baseline and Implementation phases.* *b. Inferential Statistics: A t-test for dependent samples will be conducted to compare the mean alertness scores before and after the intervention. This will determine if the 2-minute musical drill produced a statistically significant increase in student alertness ($p<0.05$).* *c. Thematic Analysis: Qualitative data from the open-ended survey questions will be coded to identify recurring themes, such as "Physical Relief," "Mental Reset," or "Audio Engagement," providing context to the numerical shifts.* *d. Triangulation: The results from the t-test will be cross-referenced with the qualitative themes and behavioral checklists to provide a holistic view of the intervention's impact on the "afternoon slump."*
VI. Ethical Considerations	**Present the plan to address the following:** • Type of participants (if involves vulnerable population):

	o The study involves students within a standard classroom setting who do not belong to vulnerable or high-risk populations. As such, no specialized protocols for protected groups are required for this specific intervention. • Risk/s to participants and researchers: o Physical risks are categorized as minimal, as the exercise routine consists of low-impact, stationary movements designed to be safe for a classroom environment. The researcher will ensure the space is clear of obstructions to prevent accidental trips or collisions during the drill. • Benefits to the participants or community o Students may experience immediate physiological benefits, such as increased blood flow and reduced lethargy, which enhance their readiness to learn Science during peak fatigue hours. The school community benefits from a scalable, evidence-based strategy to improve classroom engagement without requiring additional financial resources. • Confidentiality and handling of data o All personal identifiers will be replaced with alphanumeric codes to ensure that individual survey responses and observation scores remain anonymous. Collected data will be stored in a password-protected digital file and will be destroyed following the official conclusion of the research project. • Consent form contents o The consent form will clearly outline the purpose of the 2-minute exercise, the voluntary nature of participation, and the right of the student to opt out at any time. It will also include a specific waiver regarding the low-impact physical movements and a guarantee of data privacy. • Conflict of interest o The researcher declares no financial or personal conflicts of interest that would bias the results of this study. The primary motivation is the improvement of pedagogical practices and student wellness within the institution.
VII. Action Plan	• *Who will be involved in this plan and what would be their roles?* *a. Researcher (Proponent): Responsible for designing the 2-minute exercise routine, coordinating the MIDI audio schedule, conducting all classroom observations, and analyzing the resulting data.* *b. Subject Teacher (if not the Proponent): Assists in facilitating the transition from the exercise routine back to the Science lesson to ensure instructional flow is maintained.* *c. Students (Respondents): Participate in the daily 2-minute physical drills and provide honest feedback through the weekly post-intervention surveys.* *d. ICT / School Intercom Coordinator: Ensures the specific MIDI tracks are played through the*

school speakers at the designated time (e.g., exactly at 1:30 PM).

e. *School Principal / Supervisor: Provides the necessary administrative clearance for the use of the intercom system and monitors the study's alignment with institutional research priorities.*

f. *Parents / Guardians: Review and sign the informed consent forms regarding the physical activity component of the intervention.*

- *What is your timeline?*

Activity	Persons Involved and Roles	Dates/ Timeframe
Pre-implementation & Baseline	Research: Conducts observations using the checklist.	Week 1
Instructional Phase	Researcher: Demonstrates the drill; Students: Memorize movements.	Week 2
Intervention Implementation	Rasearcher: Monitor drills; ICT/Intercom Staff: Plays MIDI audio.	Weeks 3-6
Weekly Evaluation	Students: Complete surveys; Researcher: Tallies weekly scores.	Fridays (Weeks 3-6)

VIII. References

Carlson, J. A., Engelberg, J. K., Cain, K. L., Conway, T. L., Mignano, A. M., Bonilla, E. A., Geremia, C., & Sallis, J. F. (2015). Implementing classroom physical activity breaks: Associations with student physical activity and classroom behavior. *Preventive Medicine*, *81*, 67–72. https://doi.org/10.1016/j.ypmed.2015.08.006

Goh, T. L., Hannon, J., Webster, C., Podlog, L., & Newton, M. (2016). Effects of a take 10! classroom-based physical activity intervention on third- to fifth-grade children's on-task behavior. *Journal of Physical Activity and Health*, *13*(7), 712–718. https://doi.org/10.1123/jpah.2015-0238

MAHAR, M. T., MURPHY, S. K., ROWE, D. A., GOLDEN, J., SHIELDS, A. T., & RAEDEKE, T. D. (2006). Effects of a classroom-based program on physical activity and on-task behavior. *Medicine & Science in Sports & Exercise*, *38*(12), 2086–2094. https://doi.org/10.1249/01.mss.0000235359.16685.a3

	Ruhland, S., & Lange, K. W. (2021). Effect of classroom-based physical activity interventions on attention and on-task behavior in schoolchildren: A systematic review. *Sports Medicine and Health Science*, *3*(3), 125–133. https://doi.org/10.1016/j.smhs.2021.08.003 Schmidt, M., Benzing, V., & Kamer, M. (2016). Classroom-based physical activity breaks and children's attention: Cognitive engagement works! *Frontiers in Psychology*, *7*. https://doi.org/10.3389/fpsyg.2016.01474

Taft Ave. cor. Ayala Blvd., Ermita, Manila (+63 2) 5317 1768 loc. 570/776 | all@pnu.edu.ph

Self-Assessment Rubric for Classroom Action based Research Proposal/Concept Paper

Criteria	Exceeds Expectation 10 pts	Meets Expectation 8 pts	Nearly Meets Expectation 6 pts	Does Not Meet Expectation 2 pts	Self Rating Score	Professor's Score
Topic/Focus including (rationale, significance) related to science teaching	The topic is sharply delineated, highly significant and within the scope of an action research.	The topic is focused and significant but some ideas should be directed to suit the purpose of an action research.	The topic is broad and have little significance and within minimum scope of an action research.	The topic is totally out of scope of an action research.	10	
Integration of Knowledge on Curriculum, Instruction and Assessment on Science Teaching	The paper indicates proper application of concepts learned on curriculum, instruction and assessment of science teaching.	The paper, for the most part, indicates proper application of concepts learned on curriculum, instruction and assessment of science teaching.	The paper indicates limited application of concepts learned on curriculum, instruction and assessment of science teaching.	The paper fails to demonstrate application of concepts learned on curriculum, instruction and assessment of science teaching	10	
Graphical Conceptual Framework (Formation of Ideas)	The proposal/ concept paper presents well organized analysis and synthesis of ideas on science teaching.	The proposal/ concept paper presents organized analysis and synthesis of ideas on science teaching.	The proposal/ concept paper somehow presents organized analysis and synthesis of ideas on science teaching.	The proposal/ concept paper does not present analysis and synthesis of ideas on science	8	

				teaching.	
Research Questions/ Objectives	Questions/objectives have vital importance and clear relevance to educational practices particularly on science teaching. Questions/objectives can be answered/reached by interpreting definitive and detailed information.	Interesting questions/objectives, clear relevance to educational practices particularly on science teaching. The research questions/objectives can be answered/reached in some detail.	Questions/objectives are already addressed by a significant body of research. Questions/objectives can only be answered/reached with generalities.	Questions/objectives are unlikely to reveal new or relevant insights The research questions/objectives cannot be answered/reached.	10
Plan for Data Collection	Multiple data sources are identified and there is an attempt to triangulate data. Data sources match the problem statement and project innovation. Data collection plan is comprehensive and	Multiple data sources are identified. Data sources match the problem statement and project innovation. Data collection plan is comprehensive and doable in one school	Multiple data sources are identified. Data does not match the problem statement or innovation. Data collection plan presented needs modifications to complete or make doable in one	Few data sources are identified. Data sources do not match the problem statement or innovation. Data collection plan is insufficient.	10

	doable in one school year.	year.	school year.		
Action Plan	Action is insightfully outlined in light of several relevant educational research and best practice elements. The action plan is directly linked to a student learning goal and response is grounded in research.	Action is outlined in light of relevant educational research and best practice. The action plan is linked to a student learning goal and tentatively supported in research.	Several action steps are outlined in light relevant educational research and best practice. The action plan is loosely linked to a student learning goal and represented in some research.	Action steps are not outlined in light of relevant educational research and best practice. Action steps have been missed. The action plan might be linked to student learning but the connection is not clear in research.	10
References	References are directly related to the research study, complete, and correctly formatted.	A few references are not appropriate, incomplete and not formatted.	More than half of the references are not appropriate, incomplete and not formatted.	References are incomplete and not correctly formatted	10

Submitted to:
Prof. Dominador D. Mangao
Course Professor